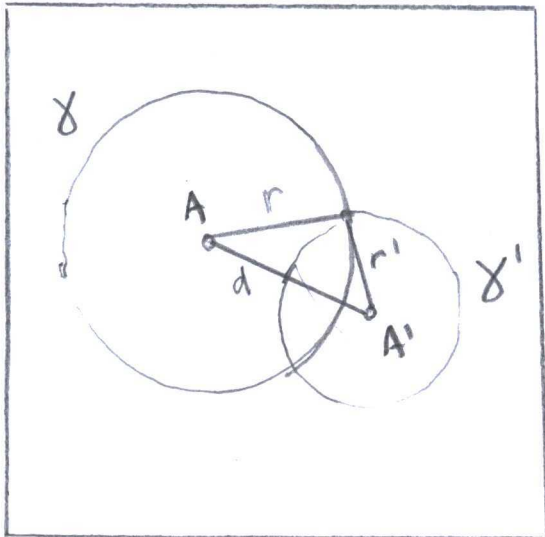


Euclidean and Non-Euclidean Geometry.

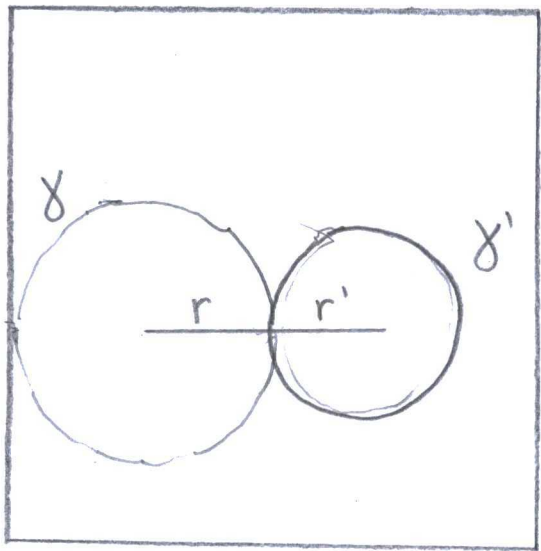
Major Exercises:

1.



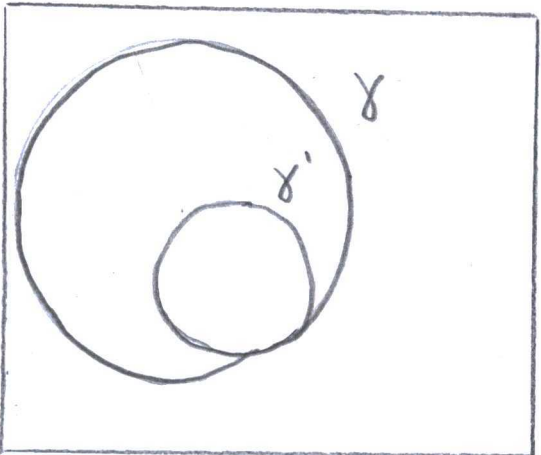
Circles intersect in two distinct points:

$$d < r + r' \\ < \frac{\gamma + \gamma'}{2\pi}$$



Circles intersect at one point:

$$d = r + r' \\ = \left(\frac{\gamma}{2\pi}\right) + \left(\frac{\gamma'}{2\pi}\right) \\ = \frac{\gamma + \gamma'}{2\pi}$$

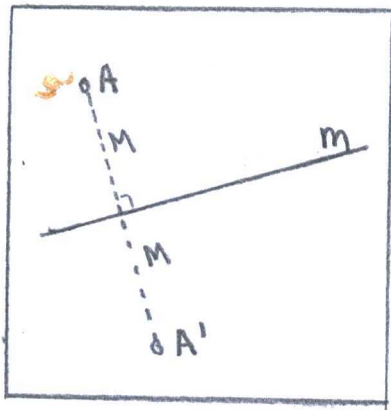


Circles are tangent:

$$d = r - r' \\ = \frac{\gamma - \gamma'}{2\pi}$$

2.

Transformation R_m is a motion:



"reflection"

Proof:

Hypothesis

Evidence

1) A point B exists on line m

Axiom I-2

2) Second point A not on m

Euclid Postulate #1

forms a line

M from A to B

3) Third point A' is on line M

Axiom B-2

4) $AB \cong A'B$

Axiom C-1

5) A point C is also on M

Axiom B-2

6) $180^\circ = \angle ABC + \angle A'BC$

(pg 19) "Definition - Supplementary

Angle"

7) $\angle ABC \cong \angle A'BC$

Proposition 3.14

(pg 122) "Motion by Felix Klein"

"a geometry as the study of those properties of figures that remain invariant under a particular

group of transformations."

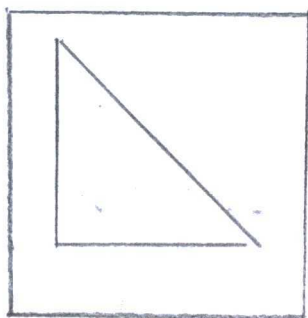
Explicit Formula for reflection (2-dimensions)
across a line:

$$y = Rx \quad \text{where} \quad R = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \\ \cong \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \theta \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

in which θ = angle of line
about reflection
 $= \tan^{-1}\left(\frac{y}{x}\right)$

$$R \text{ is also } R = \begin{pmatrix} 1 & \theta \\ \theta & 1 \end{pmatrix} \begin{pmatrix} 1 & -\theta \\ -\theta & 1 \end{pmatrix}^{-1} \\ = \frac{1}{1 + \theta^2} \begin{pmatrix} 1 - \theta^2 & 2\theta \\ 2\theta & \theta^2 - 1 \end{pmatrix} \\ = \frac{1}{1 + \left(\tan^{-1}\left(\frac{y}{x}\right)\right)^2} \begin{pmatrix} 1 - \left(\tan^{-1}\left(\frac{y}{x}\right)\right)^2 & 2\tan^{-1}\left(\frac{y}{x}\right) \\ 2\tan^{-1}\left(\frac{y}{x}\right) & \tan^{-1}\left(\frac{y}{x}\right)^2 - 1 \end{pmatrix}$$

3.

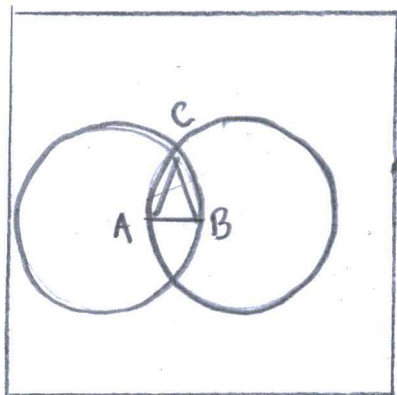


(pg 31) "Euclid I.20"

"Any two sides of a triangle
together are greater than
the third"

(pg 130) "Circle-circle Continuity Principle"

"If a circle γ has one point inside another circle γ' , then the two circles intersect in two points."



Case #1: $r + r' > d$

Proof:

Hypothesis

Evidence

1) Points A and B
reside on both
 γ and γ' with
segment AB
and circle BC γ

Euclid Postulate III

2) Points A and B
also reside on

Euclid Postulate III

BA for circle BC γ'

Euclid Postulate I

3) Point C cuts
both circles and
draws segments
CA and CB

4) $AB > AC$ and
 $AB > CB$

(pg 117) "Ordered
Field"

"Constructible
Field"

$$5) |AB| + |BC| \geq AC$$

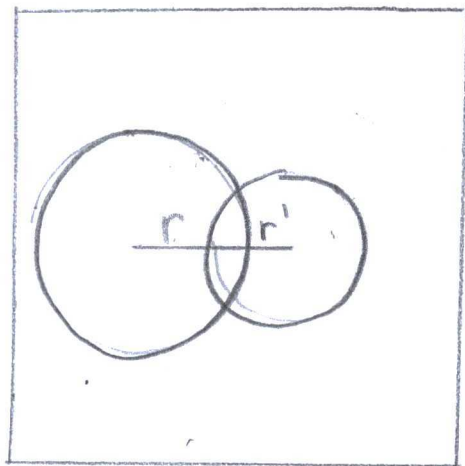
$$|AB| + |AC| \geq |BC|$$

$$|AC| + |BC| \geq |AB|$$

(pg 117) "ordered
Field"

"Constructible
Field"

Case #2: $r + d > r'$



Proof: Hypothesis

Evidence

1) Points A and B
reside on both
 γ and γ' with
segment AB
and circle $BC\gamma$

Euclid Postulate III

2) Points A and B
reside on radius
BA and describe
circle $BC\gamma'$

Euclid Postulate III

3) Point C cuts
both circles and
draws segments
CA and CB

Euclid Postulate I

4) $AB > AC$ and
 $AB > CB$

(pg 117) "Ordered Field"
"Constructable
Field"

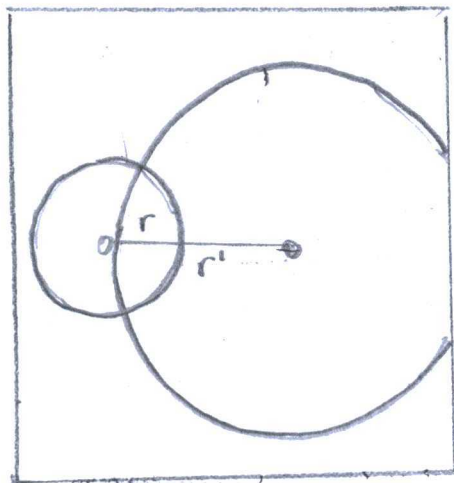
$$5) |AB| + |BC| \geq AC$$

$$\text{and } |AB| + |AC| \geq |BC|$$

$$\text{and } |AC| + |BC| \geq |AB|$$

(pg 117) "Ordered
Fields"

Case #3: $r' + d > r$



Proof: Hypothesis

Evidence

1) Points A and B
reside on γ
and γ' , respectively,
with segment AB
and circle BCX

Euclid Postulate III

2) Points A and B
also describe
radius BA for
circle BCX'

Euclid Postulate III

3) Point C cuts
both circles and
draws segments
CA and CB

Euclid Postulate I

4) $AB > AC$ and
 $AB > CB$

(pg 117) "ordered
Fields"

"Constructable
Fields"

4.

(pg 31) "Euclid I.20"

"Any two sides of a triangle together are greater than the third."

Converse to I.20:

Proof

Hypothesis

Evidence

- | | |
|---|---------------------------------|
| 1) A segment OO' exists between two triangles | Euclid Proposition <u>I</u> |
| 2) A segment ends with two points, O and O' | (Pg) "Definition-Segment" |
| 3) Segment OO' forms an equilateral triangle where the third is less than the two sides | (Euclids Proposition <u>I</u>) |
| 4) Points O and O' are centers to circles | Outcome. From Euclids Prop. 1. |

$$5) |AB| + |BC| \geq |AC|$$

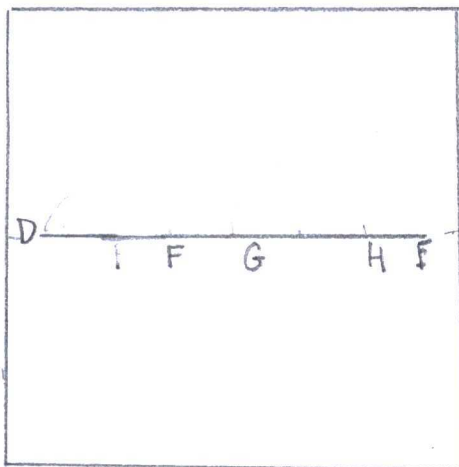
$$\text{and } |AB| + |AC| \geq |BC|$$

$$\text{and } |AC| + |BC| \geq |AB|$$

(pg 117) "ordered field"

5.

(pg 30) "Euclids I.22"



"To construct a triangle, given three sides, provided any two are greater than the third."

The obstacle about point K is circle intersection when the radius sum less than distance ($d < \overline{FG} + \overline{GH}$). The "circle-circle continuity principle" assumes intersection at two points between circles, a critical part not in "Euclids I.22."

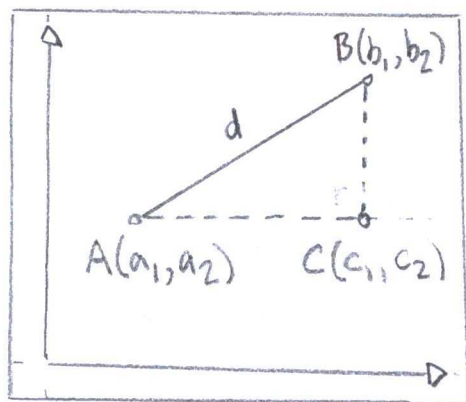
6.

(from exercise) "Taxicab Formula"

$$d^T(A, B) = |a_1 - b_1| + |a_2 - b_2|$$

a) (pg 121) "Axiom C-6 (SAS)"

"If two sides and the included angle of one



triangle are congruent, respectively, to two sides and included angle of another triangle, then the two triangles are congruent."

Note: Cartesian axes are necessary.

Axiom C-6 fails when $A=(1,0)$, $B=(0,1)$, and $C=(1,1)$. The second example relates to the image where $A=(1,1)$, $B=(2,2)$, and $C=(2,1)$. The taxicab distance is exact with horizontal and vertical sides for S.A.S. But the taxicab distance fails on the hypotenuse and a leg or base in the second example because diagonal distances.

(pg 128) "Proposition 3.22 (SSS)"

"IF $AB \cong DE$, $BC \cong EF$, and $AC \cong DF$,

then $\triangle ABC \cong \triangle DEF$."

Again the taxicab distance is greater than or equal to distance formula. A S.S.S. relationship has troubles from side lengths by taxicab distance when not vertical or horizontal.

(pg 127) "Proposition 3.17(A.S.A)"

"Given $\triangle ABC$ and $\triangle DEF$ with $\angle A \cong \angle D$, $\angle C \cong \angle F$, and $AC \cong DF$. Then $\triangle ABC \cong \triangle DEF$."

The triangles $\triangle ABC$ and $\triangle DEF$ are not 100% congruent from a taxicab distance. An equivalent side by taxicab could produce unequal areas.

Informal Axioms C-1, 2, 3, 4, 5:

(pg 119) "Congruence Axiom #1"

Segment $AB = A'B'$

(pg 120) "Congruence Axiom #2"

Segment $AB \cong CD, AB \cong EF$, then
 $CD \cong EF$ measured by taxicab.

(pg 120) "Congruence Axiom #3"

Segment $AB \cong A'B'$ from Axiom C-1,
 $BC \cong B'C'$ and $AC \cong A'C'$.

(pg 120) "Congruence Axiom #4"

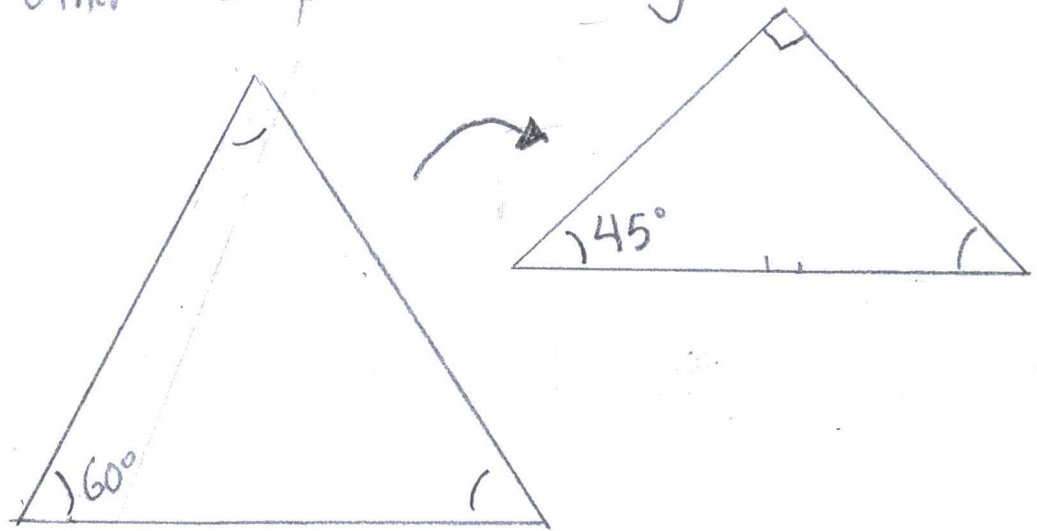
Rays in Axiom C-4 are
never consider distance,
so taxicab is okay.

(pg 121) "Congruence Axiom #5"

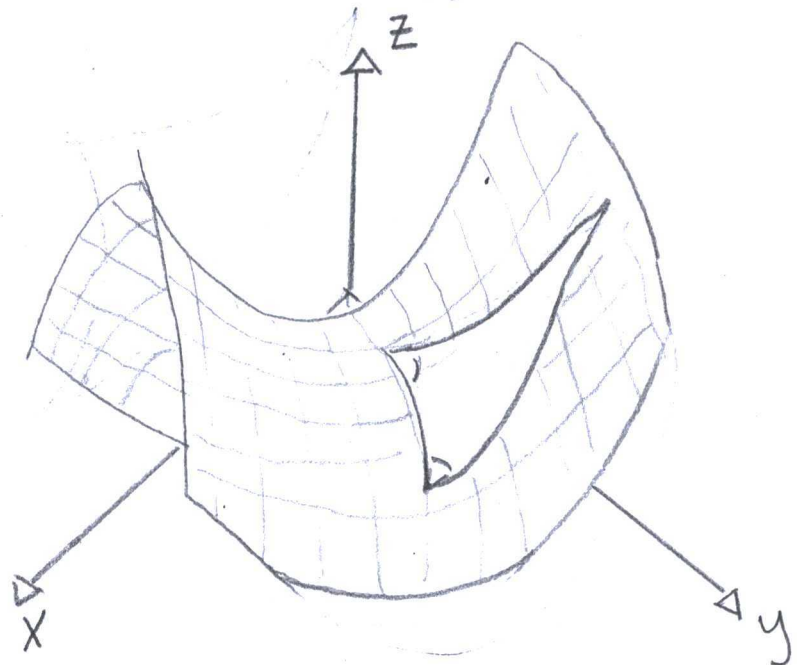
Angles are irregardless to
length and distance.

b) Equilateral Triangles
are
Isosceles:

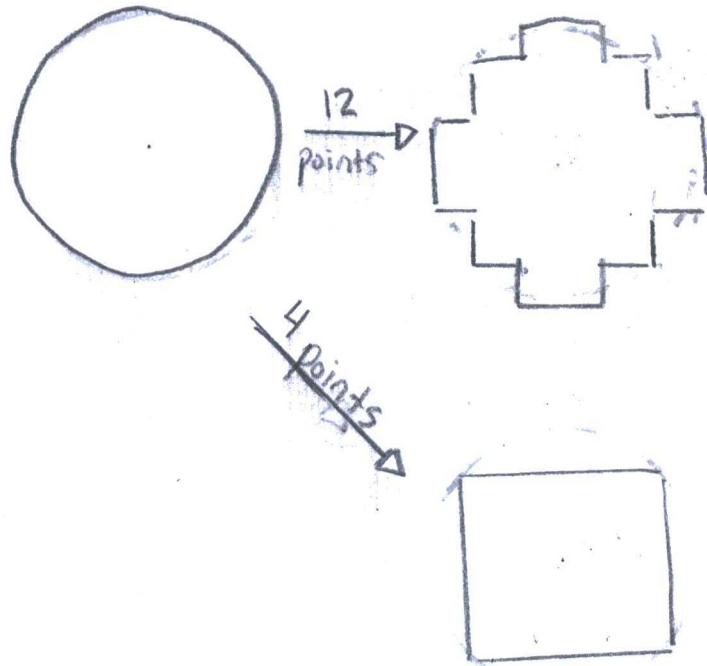
140
But by taxicab distance, produces
other shapes not congruent.



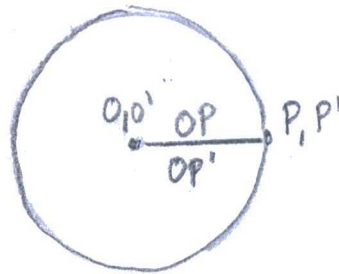
c) The only answer found, hyperbolic
triangles. Two angles are congruent
but not the opposite sides.



d) "Circle" in taxicab geometry:



e) Circles intersecting at ∞ -points:



e) Taxicab Locus:

